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Assignment-1 20L-0961 Date:

Review Questions:

R4: In P2P filesharing
peer retreiving the file is the client
peer sending the file is the server
I disagree with the statement

R6: I will use UDP to complete the transaction
as it will complete in one RTT
while TCP will take two RTT, one to setup
the connection, another to send and reply.

R10: A protocol will use handshaking if two entities
exchange control packets before sending data
SMTP uses handshake when establishing connection

R11: These protocols rely on TCP because it
guarantees delivery, strict data ordering, error-free
TCP ensures data arrives completely and transmission sequentially

R13: Web caching reduces response time for client request
this means cache is closer to client. to the same
LAN to which user host is connected.
Web caching reduce delay for all objects, even
objects that are not cached, as caching reduces
traffic on links.

R19: An organization WebServer can have some alias for hostname since type=MX is used to map mail server host name to its IP Addr

P1:

a) client sends 4 Req (HTML + 3 images) and Recv 4 Res <u>False</u>

b) Persistent → allows multiple req <u>True</u>

c) nonpersistent → new request for each resp/req <u>False</u>

d) Last-Modified: shows modification time not the Date: <u>False</u>

e) HTTP response messages omit message body <u>False</u>

P7: total RTT to retrieve IP

$RTT_1 + RTT_2 + \dots + RTT_n$

once IP address is known RTT_0 is spent to set up TCP and another RTT_0 is spent to retrieve object

hence

response time

$2RTT_0 + RTT_1 + RTT_2 + \dots + RTT_n$

Date:

P8 one Html page and 8 small objects

a) NP HTTP, no Parallel TCP ~~Choice~~

IP retrieve

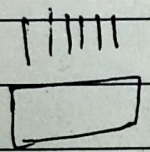
$RTT_1 + RTT_2 + \dots RTT_n + \underbrace{2RTT_0}_{\text{TCP}} + \underbrace{2 \cdot 8\, RTT_0}_{\text{8 objects}}$

$\rightarrow 18\, RTT_0 + RTT_1 + RTT_2 + \dots RTT_n$

b) NP HTP, 6 parallel TCP

$RTT_1 + RTT_2 + \dots + RTT_n + 2RTT_0 + 2 \cdot 2RTT_0$

$6\, RTT_0 + RTT_1 + RTT_2 + \dots RTT_n$



c) Persistant with pipeling

6 objects take $2RTT_0$

$RTT_1 + RTT_2 + \dots + RTT_n + 2RTT_0 + RTT_0$

$= 3\, RTT_0 + RTT_1 + RTT_2 + \dots RTT_n$

- Remaining 2 objects take $2 \cdot 2RTT_0$

Persistant w/o pipelining

$RTT_1 + RTT_2 + \dots + RTT_n + 2RTT_0 + 8RTT_0$

$= 16\, RTT_0 + RTT_1 + RTT_2 + \dots RTT_n$

P10 Link = 10m $R = 150$ b/s $L = 100{,}000$ b

control packets = 200 b T_p = prop delay

parallel downloads, non persistant

$\downarrow (10)$

$\frac{150}{10} \rightarrow 15$ b/s

first $\left(3 \times \left(\frac{200}{150} + T_p\right) + \left(\frac{100{,}000}{150} + T_p\right)\right) +$

10 parallel $\left(3 \times \left(\frac{200}{150} + T_p\right) + \left(\frac{100{,}000}{15} + T_p\right)\right)$

$= 7377 + 8T_p$ sec

Persistent HTTP

$$\left(3\times\left(\frac{200}{150}+T_p\right)+\frac{100{,}000}{150}+T_p\right)+$$

$$10\left(\frac{200}{150}+T_p+\frac{100{,}000}{150}+T_p\right)$$

$= 7351 + 24\,T_p$ sec

Now if we assume $T_p = \frac{10}{300\times10^6} = 0.03$ msec

hence I have ignored T_p as it is vv small

→ It can be concluded that persistent HTTP is only faster than by 1%

P11: a) Yes Bob has more connections
he gets larger share of bandwidth

b) they will be beneficial as Bob will use parallel downloads, if he does not then he will get less bandwidth

P20: We can periodically take snapshot of DNS records in local DNS servers.
The webserver that appears most frequently in DNS caches is most popular server.
If more users are interested in the webserver then DNS requests for that server are sent frequently by users
That webserver will also appear in DNS caches more frequently.